

Visualizing Generalized Pentagonal Numbers

• Standard Pentagons ($k \geq 0$) $\frac{k(3k-1)}{2}$
• Generalized Addition ($k < 0$, visual $g_{-k} = g_k + k$)

$k = 0$

$k = 1$

g_1

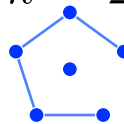


$k = -1$

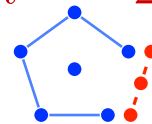


$g_{-1} = g_1 + 1$

$k = 2$

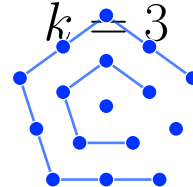


$k = -2$



$g_{-2} = g_2 + 2$

$k = 3$



Calculation:

$$\frac{1(3 \cdot 1 - 1)}{2} =$$

$$\frac{1(2 \cdot (-1) - 1)}{2} =$$

Calculation:

$$\frac{1(2 \cdot (-1) - 1)}{2} =$$

$$\frac{(-1)(-4 \cdot 2 - 1)}{2} =$$

Calculation:

$$\frac{(-1)(-4 \cdot 2 - 1)}{2} =$$

$$\frac{2(5 \cdot (-2) - 1)}{2} =$$

Calculation:

$$\frac{2(5 \cdot (-2) - 1)}{2} =$$

$$\frac{(-2)(-7 \cdot 3 - 1)}{2} =$$

Calculation:

$$\frac{(-2)(-7 \cdot 3 - 1)}{2} =$$

$$\frac{3(8)}{2} =$$

Calculation:

$$\frac{0(0-1)}{2} = 0$$

1

2

5

7

12

0